

Ansible Cheat Sheet

Key Ansible components

Playbooks

YAML files that contain all the tasks you want to run against your hosts.

Modules

Can perform tasks such as copying a file or installing an application.

Inventory

Collection of all the hosts you want to run your Playbooks against. This can be static or dynamic and be passed into your Ansible commands using the `-i` flag.

Variables

You can store specific values to be used throughout your playbooks. These can also be passed in via Ansible Commands or defined in your playbooks or inventory.

Ansible facts

A way to retrieve information from the system you are running your playbook against.

Handlers

Specific tasks in a playbook that can trigger a handler, called during the end of a playbook.

Roles

Allow you to take repetitive tasks and store them in a single role, enabling reusability and structure for managing your Ansible environment.

Rebooting and shutting down Hosts

```
ansible -i ~/myinventory all -m reboot
```

Reboot all hosts

```
ansible -i ~/myinventory all -m command -a "/sbin/shutdown -h now"
```

Shutdown all hosts using command module

Mounting disks operations

```
ansible -i ~/myinventory all -m mount -a "path=/mnt/mydisk src=/dev/sdb1 fstype=ext4 state=mounted"
```

Mount a filesystem

```
ansible -i ~/myinventory all -m mount -a "path=/mnt/mydisk state=unmounted"
```

Unmount a filesystem

Using Cron module to manage Cron jobs

```
ansible -i ~/myinventory all -m cron -a "name='backup' minute=0 hour=2 job='/usr/local/bin/backup.sh'"
```

Create a cron job

```
ansible -i ~/myinventory all -m cron -a "name='backup' state=absent"
```

Remove a cron job

Basic Ansible playbook commands

```
ansible-playbook myplaybook.yml -i ~/myinventory
```

Execute a playbook

```
ansible-playbook myplaybook.yml -i ~/myinventory --check
```

Dry-run a playbook before fully executing

```
ansible-playbook myplaybook.yml -i ~/myinventory --limit web
```

Specify inventory group on the command line level if the playbook is targeting all

```
ansible-playbook playbook.yml -v
```

Run playbook with verbose for troubleshooting purposes

```
ansible-playbook playbook.yml -vv
```

Extra verbose

```
ansible-playbook playbook.yml -vvv
```

Even more verbose

Performing file operations on Hosts with Ansible using copy, fetch and file modules

```
ansible -i ~/myinventory all -m copy -a "src=/path/to/source dest=/path/to/destination"
```

Copy a file from Control Node to Host

```
ansible -i ~/myinventory all -m copy -a "src=/path/to/source dest=/path/to/destination" --check
```

Run a Copy file check to validate action before running

```
ansible -i ~/myinventory all -m fetch -a "src=/remote/path dest=/local/path"
```

Fetch a file from Host to the Control Node

```
ansible -i ~/myinventory all -m file -a "path=/path/to/directory state=directory mode=0755"
```

Create a directory on Hosts

```
ansible -i ~/myinventory all -m file -a "path=/path/to/file state=absent"
```

Remove a file or directory from Hosts

```
ansible -i ~/myinventory all -m file -a "src=/path/to/source dest=/path/to/symlink state=link"
```

Create a Symbolic Link on Hosts

Managing users and groups using the user module

```
ansible -i ~/myinventory all -m user -a "name='username' password='<hashed_password>' state=present" --become --ask-become-pass
```

Create a User. The password value must be hashed in order to run this command. You can use `openssl` or another tool to hash your password.

```
ansible -i ~/myinventory all -m user -a "name=username state=absent"
```

Add a User to Group

```
ansible -i ~/myinventory all -m user -a "name=username groups=admin append=yes"
```

Add a User to Group

Installing applications to Hosts

```
ansible -i ~/myinventory all -m apt -a "name=nginx state=present"
```

Install a Package using apt package manager

```
ansible -i ~/myinventory all -m yum -a "name=nginx state=present"
```

Install a Package using yum package manager

```
ansible -i ~/myinventory all -m apt -a "name=httpd state=absent"
```

Remove a Package using apt package manager

```
ansible -i ~/myinventory all -m yum -a "name=httpd state=absent"
```

Remove a Package using yum package manager

```
ansible -i ~/myinventory all -m apt -a "upgrade=dist"
```

Upgrade all Packages using apt package manager

```
ansible -i ~/myinventory all -m yum -a "upgrade=dist"
```

Upgrade all Packages using yum package manager

Managing services on hosts using service and systemd modules

```
ansible -i ~/myinventory all -m service -a "name=nginx state=started"
```

Start a service

```
ansible -i ~/myinventory all -m service -a "name=nginx state=stopped"
```

Stop a service

```
ansible -i ~/myinventory all -m systemd -a "name=nginx state=restarted"
```

Restart a service with systemd

```
ansible -i ~/myinventory all -m systemd -a "name=nginx enabled=yes"
```

Enable a service with systemd

Ansible inventory commands

```
ansible-inventory -i ~/myinventory --list
```

Display your inventory file in a JSON format

```
ansible-inventory -i ~/myinventory --list -y
```

Display your inventory file in a YAML format

```
ansible-inventory -i ~/myinventory --graph
```

Get a Tree structure of your inventory file

```
ansible-inventory -i ~/myinventory --host hostname
```

Find out specific information about a host in an inventory file

Useful flags to use with Ansible playbooks

```
ansible-playbook playbook.yml -i ~/myinventory --extra-vars "key1=value1 key2=value2"
```

Pass in extra variables from the command line to your playbook

```
ansible-playbook playbook.yml -i ~/myinventory --start-at-task "install nginx package"
```

Start from a specific task

```
ansible-playbook playbook.yml -i ~/myinventory -u <username>
```

Use specific user to connect to remote hosts

```
ansible-playbook playbook.yml -i ~/myinventory --ask-pass
```

Get prompted to enter password to SSH into Hosts instead of using SSH key pair

```
ansible-playbook playbook.yml -i ~/myinventory --ask-become-pass
```

Get prompted to escalate to sudo rights

```
ansible-playbook playbook.yml -i ~/myinventory --list-hosts
```

List all hosts in inventory that will be affected by this playbook

```
ansible-playbook playbook.yml -i ~/myinventory --step
```

Run your playbook step by step

```
ansible-playbook playbook.yml -i ~/myinventory --syntax-check
```

Check for syntax errors on the playbook before running

Ansible Galaxy commands

```
ansible-galaxy list --role-path ~/roles
```

List all Ansible roles in your local roles directory

```
ansible-galaxy collection list
```

List all Ansible roles in the Community managed roles directory

```
ansible-galaxy role list
```

List all Ansible roles installed on your Control Node

```
ansible-galaxy info /path/to/local/role
```

View information of your custom role

```
ansible-galaxy info grafana
```

View information of a galaxy role

```
ansible-galaxy search <role_name> or ansible-galaxy search apache
```

Search for a role by a specific topic or key in Ansible Galaxy

```
ansible-galaxy remove grafana
```

Remove a role from your system

Ansible Vault commands

```
ansible-vault create secrets.yml
```

Create an encrypted file to store secrets

```
ansible-playbook playbook.yml --ask-vault-pass
```

Enter password

```
ansible-playbook playbook.yml --vault-password-file /path/to/vault_password
```

Pass in the Vault password file

Managing Ansible vault files

```
ansible-vault encrypt vars.yml
```

Encrypt existing file

```
ansible-vault decrypt vars.yml
```

Decrypt encrypted file

```
ansible-vault view secrets.yml
```

View encrypted file

```
ansible-vault edit secrets.yml
```

Edit encrypted file

```
ansible-vault rekey secrets.yml
```

Change password for encrypted file